

SNF-Bench: Separating Static Drift from Natural Flow in Long-Horizon Fixed-Camera Video Generation

Supplementary Material

This supplement records the material supporting the main paper without being required to follow its argument: the region-partition procedure and its sensitivity study, complete audit tables for both tracks at every horizon, measurement coverage, protocols for the two complementary perceptual checks, and additional qualitative material. The text-conditioned and image-conditioned tracks are reported here at equal depth. Section numbering is independent of the main paper.

1. Formal Factor Definitions

The definitions below are those referenced from the main paper’s protocol section. Notation follows it: I_t is frame t of a T -frame sequence, Ω_{static} and Ω_{flow} are the region masks, $\mathbf{u}_t$ is the optical flow from I_t to I_{t+1} , and $\mathcal{E}$ and $\mathcal{L}$ are the early and late windows.

Feature-aligned Background Drift. For matched ORB pairs $\mathcal{M}_t$ between the first frame and frame t ,

$$D_{\text{feat}}(t) = \text{median}_{(p,q) \in \mathcal{M}_t} \|p - q\|_2, \\ \text{fBD} = \frac{100}{d} \cdot \frac{1}{|\mathcal{L}|} \sum_{t \in \mathcal{L}} D_{\text{feat}}(t) \quad (\downarrow), \quad (1)$$

with d the frame diagonal, so fBD is a percentage of image size.

Normalised Background Flow.

$$\text{NBF} = \frac{10^3}{W \Delta t} \cdot \frac{1}{T-1} \sum_{t=1}^{T-1} \frac{1}{|\Omega_{\text{static}}|} \sum_{\mathbf{x} \in \Omega_{\text{static}}} \|\mathbf{u}_t(\mathbf{x})\|_2 \quad (\downarrow), \quad (2)$$

with W the frame width and Δt the interval of the sampled pairs actually used for flow estimation, so the quantity is independent of both the native frame rate and the sampling policy.

Global-motion compensation. With $T_t(\mathbf{x}) = s_t R_t \mathbf{x} + \mathbf{b}_t$ the robust similarity transform fitted to static-region correspondences and $\mathbf{d}_t(\mathbf{x}) = T_t(\mathbf{x}) - \mathbf{x}$ the displacement field it induces,

$$\tilde{\mathbf{u}}_t(\mathbf{x}) = \mathbf{u}_t(\mathbf{x}) - \mathbf{d}_t(\mathbf{x}). \quad (3)$$

Motion-Compensated Foreground Flow and Flow Persistence.

$$\text{MCFF}(\mathcal{W}) = \frac{1}{|\mathcal{W}|} \sum_{t \in \mathcal{W}} \frac{1}{|\Omega_{\text{flow}}|} \sum_{\mathbf{x} \in \Omega_{\text{flow}}} \|\tilde{\mathbf{u}}_t(\mathbf{x})\|_2 \quad (\uparrow), \quad (4)$$

$$\text{FP} = \min \left(\frac{\text{MCFF}(\mathcal{L})}{\text{MCFF}(\mathcal{E}) + \epsilon}, c \right) \quad (\uparrow), \quad (5)$$

clipped at a fixed c so that a vanishing early window cannot produce an unbounded ratio.

Drift leakage and attenuation. With F_{static} the mean flow magnitude over Ω_{static} and $F_{\text{raw}}, F_{\text{comp}}$ the mean raw and compensated magnitudes over Ω_{flow} , all in the late window,

$$\text{DLR} = \frac{F_{\text{static}}}{F_{\text{raw}} + \epsilon} \quad (\downarrow), \quad \text{DAR} = 1 - \frac{F_{\text{comp}}}{F_{\text{raw}} + \epsilon} \quad (\downarrow). \quad (6)$$

DLR is not a fraction and is deliberately unbounded: $\text{DLR} > 1$ indicates static support moving more than the region intended to move. DAR is stored signed and reported clipped to $[0, 1]$.

2. Region Partition

Derivation. For every generated sequence we average optical-flow magnitude [8] over the leading 12% of sampled frame pairs and threshold it by Otsu’s method [4] into a dynamic region and its complement. The static region is then reduced by a fixed morphological erosion and a 4% frame border discarded; together these form the ignored transition band, so boundary pixels—where the partition is least certain and flow least reliable—contribute to neither region. The same procedure applies to both tracks.

Why the estimate is restricted to an early window. Deriving a partition from a sequence’s entire motion field would be circular, since accumulated background drift would itself be read as evidence that the affected region is dynamic. Restricting the estimate to the early window breaks that circularity in the direction that matters: at the 60s horizon the partition is fixed from the first 7.1s, before long-horizon drift has accumulated, and the metrics are evaluated over the remainder.

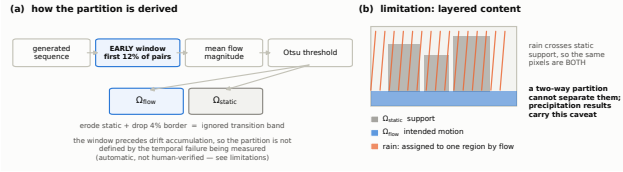


Figure 1. Region-partition protocol. (a) The partition is derived automatically from early-window flow, with an ignored transition band at the boundary. (b) A two-way partition cannot represent layered content such as rain crossing static support, which is the scope limit stated in the main paper.

Limitations. The partition is obtained automatically rather than by annotation, so no inter-annotator agreement is reported, and for the image-conditioned track it is derived per generated sequence rather than once from the shared source image. The early window is a weaker guarantee than a first-frame annotation: a system that drifts within the first 12% of its rollout can influence its own partition. We therefore treat partitions as measured quantities and report the consequence directly—every measurement is recomputed under systematic erosion and dilation of the boundary, with the response in Table 1.

3. Validation Protocol

Geometric severity is specified in a common physical unit—the mean pixel displacement a corruption induces over the frame by the end of the rollout—and each family’s parameter is derived from it using the frame geometry: a translation of d pixels, a rotation of $d/\bar{r}$ radians, or a zoom of $1 + d/\bar{r}$, where $\bar{r}$ is the mean distance from the frame centre. This is what makes the geometric families comparable. A parameter in degrees is not commensurate with one in pixels: at the evaluation resolution one degree of rotation induces roughly 2.7 px of mean displacement, so families compared by their native parameters differ in injected severity by more than an order of magnitude, and a factor can appear non-monotone simply because the corruption was too weak to clear the clip’s own baseline drift.

4. Generation Configuration

Every audited system was run in its released configuration. What the run manifest records is reported below; what it does not record is stated rather than reconstructed. Per-method sampler settings—denoising steps, chunk length, guidance scale, cache policy—were not captured at generation time, and we do not infer them after the fact. This is the weakest point in the reproducibility of the audit, and it constrains what the

Factor	target	ρ	ρ_{rot}	off-target	admitted
fBD	drift	+1.00	+0.70	14%	✓
NBF	drift	+1.00	+0.60	6%	✓
MCFF-L	freeze	+1.00	n/a	14%	✓
FP	freeze	+0.80	n/a	21%	✓
DLR	drift	+1.00	-0.80	5%	✓
DAR	drift	+0.90	+1.00	18%	✓
VBench DD	drift	+0.70	+0.70	2%	✓

Table 1. Mechanistic validation and factor admission. Each factor is scored on the corruption its definition says it should detect: accumulating global drift for the static-fidelity and drift diagnostics, progressive freezing for the persistence factors. ρ is Spearman correlation against injected severity, signed so that a correct response is positive; ρ_{rot} repeats it for rotational drift, the harder case. off-target is the largest relative excursion under corruptions the factor should ignore—photometric drift and perturbation of the region partition—so a small value indicates selectivity. Admission requires $\rho \geq 0.7$ and off-target response below 25%. Twelve reference clips per severity level.

results support: they compare the systems as released and run here, not the space of settings each admits.

5. Measurement Coverage

Coverage is counted from valid per-video measurements rather than from the presence of a results file, so a run that terminated early is reported as partial. Separately, fBD is feature-based [6] and declines to report where too few repeatable keypoints survive; those abstentions are tabulated rather than folded silently into a smaller n .

6. Complete Audit, Both Tracks

The main paper reports the text-conditioned track at 60s, the tier whose sample supports method-level separation. Both tracks appear here at every horizon, with uncertainty from a percentile bootstrap over items [1]. For the image-conditioned track, systems marked matched run under a common long-horizon wrapper rather than their released configuration, so their scores are read as deployment sensitivity rather than as the published method’s performance.

7. Aggregation Sensitivity

The evaluation set is dominated by one flow medium, so the audit must be shown not to be a report on that medium alone. The main paper aggregates by averaging over prompts. The alternative, weighting each scene category equally, is not preferable here: at the principal horizon three of the five populated categories contain two prompts each, so a macro-average would give a two-prompt cell the same influence as a fourteen-

System	setting	resolution	fps	horizons
CausVid	native	832×480	16	5s, 60s, 120s, 240s
Self-Forcing	native	832×480	16	5s, 60s, 120s, 240s
Infinite-Forcing	native	832×480	16	5s, 60s, 120s, 240s
Rolling-Forcing	native	832×480	16	5s, 60s, 120s, 240s
Reward-Forcing	native	832×480	16	5s, 60s, 120s, 240s
LongLive	native	832×480	16	5s, 60s, 120s, 240s
Causal-Forcing	native	832×480	16	5s, 60s, 120s, 240s

Table 2. Recorded generation configuration. Every audited system was run in its released configuration and produced output at the same resolution and frame rate over the same horizons, so no comparison in this paper is confounded by output format. Per-method sampler settings—denoising steps, chunk length, guidance scale, cache policy—were not captured in the run manifest and are therefore not listed here; this is a reproducibility gap we state rather than fill by inference.

track	horizon	measured	fBD abstained	rate
I2V	120s	400	0	0.0%
I2V	240s	65	0	0.0%
I2V	5s	180	0	0.0%
I2V	60s	630	7	1.1%
T2V	120s	42	0	0.0%
T2V	240s	28	0	0.0%
T2V	5s	48	0	0.0%
T2V	60s	185	0	0.0%

Table 3. Measurement coverage and fBD abstention. fBD is feature-based and declines to report when too few repeatable keypoints survive in the static region. Every abstention in this benchmark occurs on a single desert dust-storm scene, whose flat, dust-obscured ground offers no stable structure to track, and it occurs there for every system alike—so it reflects the scene, not a method. The remaining factors are reported for those clips as usual; only fBD is withheld.

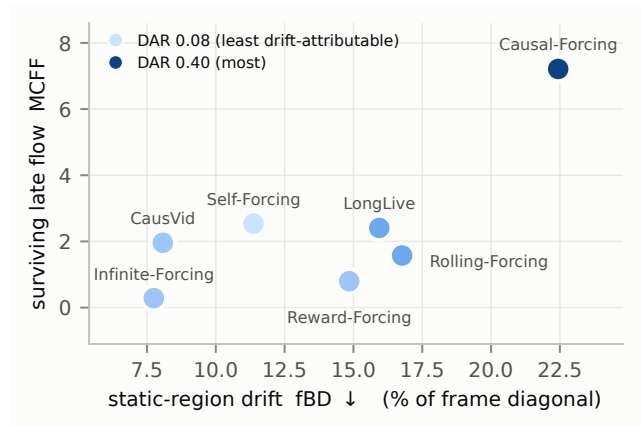


Figure 2. Operating regimes under native deployment. Each system in the static-drift-surviving-flow plane. Colour encodes the relative attenuation of measured dynamic-region flow after global-motion compensation, not a share of it attributable to drift: the quantity is signed, and values are shown clipped to $[0, 1]$. The view exposes trade-offs; it is not a leaderboard.

134 prompt one and inflate variance in exchange for nomi-
 135 nal balance. We therefore report both and state what
 136 changes.

137 The best- and worst-ranked systems are the same
 138 under either aggregation, so the contrast the paper
 139 draws does not depend on the choice. Intermediate
 140 ranks do move, which is why no ordinal claim is made
 141 about the middle of the field.

142 8. Whole-Frame and SNF-Bench Rank Corre-
 143 lation

144 9. Deployment Sensitivity

145 10. Signed Drift Attenuation

146 Drift attenuation is stored signed and reported clipped
 147 to $[0, 1]$. Negative values are informative rather than
 148 anomalous: where local flow opposes the estimated
 149 global field, compensation increases the measured mag-

nitude. We report their incidence rather than suppress-
 ing it.

11. Evaluation-Set Composition

12. Preliminary Perceptual Validation of the
 Factor Axes

This section asks one question: do independent judges
 order clip pairs the way the factors do? It is metric
 validation, not method preference. Nothing here ranks
 systems, and no result from it enters any factor, table
 or figure elsewhere in this work. Framed that way a
 small sample is survivable, because the object under
 test is a measurement instrument rather than a leader-
 board.

Method	status	setting	5s	60s	120s	240s
CausVid	public	native	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B
Self-Forcing	public	native	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B
Infinite-Forcing	public	native	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B
Rolling-Forcing	public	native	V6 T6 B	V24 T24 B	V6 T6 B	V4 T4 B
Reward-Forcing	public	native	V12 T12 B	V23 T23 B	V6 T6 B	V4 T4 B
LongLive	public	native	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B
Causal-Forcing	public	native	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B
Method	status	setting	5s	60s	120s	240s
Causal-Forcing++ (2-step)	public	matched	V10 T10 X B	V30 T30 X B	V20 T20 X B	V5 T5 X B
Causal-Forcing++ (1-step)	public	matched	V10 T10 X B	V30 T29/30 X B	V20 T20 X B	V5 T5 X B
Causal-Forcing (frame-wise)	public	matched	V10 T10 X B	V30 T30 X B	V20 T20 X B	V5 T5 X B
Causal-Forcing++ (2-step, frame-wise)	public	native	V10 T10 X B	V30 T30 X B	V20 T20 X B	V5 T5 X B
Self-Forcing	public	matched	V10 T10 X B	V30 T30 X B	V20 T20 X B	V5 T5 X B
CausVid	public	matched	V10 T10 X B	V30 T29/30 X B	V20 T20 X B	V5 T5 X B
Wan2.1-I2V-14B-480P	public	native	V10 T10 X B	.	.	.
Wan2.2-I2V-A14B	public	native	V10 T10 X B	.	.	.
LTX-Video 13B-0.9.8-distilled	public	native	V10 T10 X B	.	.	.
Causal-Forcing++ (1-step, frame-wise)	public	native	.	V30 T30	V20 T20	.

Table 4. Asset and score coverage for the audited public systems. Tn/N marks entries where a metrics file exists but only n of N videos hold usable measurements.

163 Questions. One forced choice per axis, with an ex-
 164 plicit third option: static stability—in which clip does
 165 the background stay more fixed in place—and motion
 166 persistence—in which clip does the intended motion
 167 keep going rather than fade or freeze. Judges are never
 168 asked which output is better, or which looks more nat-
 169 ural. “Better” conflates the two failures the benchmark
 170 separates, and naturalness is outside the stated scope,
 171 so either question would produce a result the paper
 172 could not own.

173 Pairs. A pair is two systems’ outputs on the same
 174 prompt at the same horizon, never across prompts.
 175 Pairs are stratified by the relative gap in the axis’s own
 176 factor: clear-gap pairs, where the factor separates the
 177 two outputs, carry the agreement claim, while near-tie
 178 pairs test the opposite property—that where the fac-
 179 tors do not separate, judges also cannot tell the clips
 180 apart. Presentation side is randomised per trial and re-
 181 solved back to a system identity only after the response
 182 is recorded; system names are never shown.

183 Statistics. The unit of analysis is the pair, majority-
 184 voted across raters, and intervals are percentile boot-
 185 strap resampling pairs. Responses are clustered inside
 186 rater and inside pair, so an interval computed over re-
 187 sponses would claim precision the design does not have.
 188 We report inter-rater agreement so the reader can see
 189 whether the raters are signal or noise, and we run no
 190 significance test the sample cannot power: the claim is
 191 descriptive agreement, and saying so plainly is prefer-
 192 able to a p -value that would not survive scrutiny.

193 Outcome of the human arm: the instrument works,
 194 agreement is not yet estimable. Eight raters each

completed all 24 trials, four of them independent of the
 authors; the remaining four were pilot sessions and are
 reported separately rather than pooled. Engagement
 was genuine—median deliberation ran from 23 to 233
 seconds per trial—and the “can’t tell” option was used
 deliberately, on 30% of responses, rather than avoided.

The agreement estimate, however, does not survive
 its own error bars. Majority voting across raters leaves
 three decided pairs on the static axis and five on the
 persistence axis, and the corresponding intervals span
 most of the unit range. Inter-rater agreement is low
 ($\alpha = 0.18$ and 0.24 on the two axes), and the direc-
 tion of the estimate reverses depending on which rater
 subset is used. We therefore report the counts and the
 interval rather than a headline percentage, and we do
 not claim that perceptual agreement has been estab-
 lished.

The cause is a property of the pair set rather than of
 the raters, and it is measurable. The pairs were strat-
 ified by the gap in each axis’s own factor but were not
 isolated against the other axis. On the static ques-
 tion the median relative gap in the target factor is
 0.29 while the median gap in the persistence factor is
 0.58: the clips shown differ about twice as much on the
 axis the rater was told to ignore as on the one being
 asked about, and on half the pairs both axes differ sub-
 stantially. A rater seeing one clip drift and the other
 stall cannot report which axis drove the answer, which
 is exactly the condition that produces low inter-rater
 agreement and near-chance ordering. A powered study
 therefore requires axis-isolated pairs—large gap in the
 target factor, small gap in the other—and that con-
 straint, not rater count, is the first thing to fix.

Method	fBD↓	NBF↓	MCFF-E	MCFF-L↑	FP↑	DLR↓	DAR↓
5s horizon ($n = 6$)							
CausVid	2.08	6.56	1.44	2.91	1.24	0.26	0.07
Self-Forcing	0.68	10.8	7.70	4.35	0.76	0.49	0.03
Infinite-Forcing	0.24	2.32	1.23	0.89	1.08	0.21	0.09
Rolling-Forcing	3.51	12.1	3.10	3.69	1.32	0.48	0.17
Reward-Forcing	1.10	5.21	0.63	0.58	1.37	0.44	0.28
LongLive	1.91	7.31	1.25	2.27	1.16	0.59	0.32
Causal-Forcing	9.55	31.0	5.32	3.01	1.13	0.62	0.37
60s horizon ($n = 23$)							
CausVid	8.08	9.27	2.87	1.96	0.67	0.45	0.14
Self-Forcing	11.4	12.7	10.4	2.54	0.44	0.50	0.14
Infinite-Forcing	7.75	3.65	1.17	0.29	0.63	0.55	0.14
Rolling-Forcing	16.8	11.2	3.18	1.57	0.75	0.54	0.18
Reward-Forcing	14.8	8.51	3.81	0.80	0.49	0.60	0.17
LongLive	15.9	13.9	6.46	2.41	0.75	0.59	0.17
Causal-Forcing	22.4	87.6	9.84	7.21	0.83	0.94	0.43
120s horizon ($n = 6$)							
CausVid	6.15	3.42	0.36	0.52	1.06	0.41	0.13
Self-Forcing	21.2	6.80	4.85	0.82	0.85	0.75	0.15
Infinite-Forcing	16.1	2.60	0.50	0.32	0.64	0.57	0.21
Rolling-Forcing	23.8	10.6	3.33	0.77	0.38	0.82	0.22
Reward-Forcing	20.0	11.6	2.05	0.69	0.39	0.70	0.12
LongLive	22.5	9.66	5.10	1.19	0.81	0.84	0.23
Causal-Forcing	21.8	130	10.5	10.7	0.90	0.81	0.50
240s horizon ($n = 4$)							
CausVid	7.23	13.4	0.42	0.87	0.99	0.85	0.08
Self-Forcing	20.0	17.4	3.09	0.50	0.15	2.26	0.20
Infinite-Forcing	20.0	4.15	0.33	0.23	1.02	1.08	0.15
Rolling-Forcing	23.7	15.4	2.15	0.97	0.46	0.78	0.18
Reward-Forcing	26.4	11.4	1.09	3.55	1.53	0.50	0.07
LongLive	23.1	11.8	1.52	1.32	1.20	1.38	0.31
Causal-Forcing	18.9	102	4.72	6.90	1.21	0.73	0.58

Table 5. Complete T2V audit, all four horizons. The main paper reports the 60s and 240s panels; the 5s tier is an initialisation check and 120s an intermediate diagnostic. Every row is computed under the same frozen specification, so panels are comparable within a horizon; prompt sets differ across horizons, so columns are not comparable between panels. All systems are evaluated in the native setting. n is stated per horizon and is common to every row of that panel.

228 Outcome of the vision-language arm: negative, with a
 229 located cause. The same pairs and the same wording
 230 were put to an open-weight judge [5] at temperature
 231 zero, each pair asked in both presentation orders. Two
 232 controls precede any reading of its answers. Shown one
 233 clip against itself, where the only correct answer is that
 234 they are the same, it answered correctly on six of six.
 235 Shown a frozen single frame against a real sequence—
 236 the largest difference available on the motion axis—its
 237 verdict followed the swapped clip on only two of six.
 238 Presentation bias is absent: over 48 queries the two
 239 slots were chosen 8 and 12 times. The judge is therefore
 240 well behaved and still cannot do the task: it answered
 241 “indistinguishable” on most pairs the factors separate
 242 clearly, leaving one and two decided pairs on the two

axes, which is far too few to support an agreement rate.
 We report none.

The controls locate the cause in the judge rather
 than in the factors or the prompt. A model that can-
 not separate a still image from moving water is not
 positioned to separate two rollouts that differ subtly.
 This says that an open-weight judge at this scale can-
 not presently substitute for human raters on these axes.
 It does not say that stronger judges would fail, and we
 make no claim about them: contemporary generation
 benchmarks that report successful automated judging
 use frontier proprietary models [2, 7] or scorers fine-
 tuned for the task [3], and we have no evidence either
 way about those here.

Method	setting	fBD↓	NBF↓	MCFF-E	MCFF-L↑	FP↑	DLR↓	DAR↓
5s horizon ($n = 10$)								
Causal-Forcing++ (2-step)	matched	3.69	10.0	1.06	0.61	0.78	0.67	0.32
Causal-Forcing++ (1-step)	matched	5.28	11.4	1.05	0.47	0.64	0.66	0.29
Causal-Forcing (frame-wise)	matched	16.1	45.3	8.48	2.79	0.43	0.68	0.29
Causal-Forcing++ (2-step, frame-wise)	native	1.22	10.6	2.23	0.78	0.47	0.68	0.20
Self-Forcing	matched	9.66	5.74	0.69	0.26	0.70	0.74	0.21
CausVid	matched	15.2	10.8	0.76	0.87	0.98	1.04	0.10
Wan2.1-I2V-14B-480P	native	5.01	21.3	6.67	5.58	1.09	0.52	0.18
Wan2.2-I2V-A14B	native	6.31	25.3	4.33	9.09	0.95	0.50	0.27
LTX-Video 13B-0.9.8-distilled	native	6.65	17.8	3.59	3.48	0.92	0.39	0.24
60s horizon ($n = 30$)								
Causal-Forcing++ (2-step)	matched	9.18	5.97	0.85	0.60	0.86	0.67	0.25
Causal-Forcing++ (1-step)	matched	5.34	7.05	0.86	0.57	0.77	0.64	0.28
Causal-Forcing (frame-wise)	matched	21.1	36.4	2.80	2.69	0.89	0.80	0.35
Causal-Forcing++ (2-step, frame-wise)	native	9.94	6.91	1.30	0.78	1.02	0.75	0.17
Self-Forcing	matched	17.3	5.52	1.50	0.69	0.89	0.67	0.18
CausVid	matched	19.9	3.31	0.58	0.41	0.81	0.57	0.14
Causal-Forcing++ (1-step, frame-wise)	native	10.4	5.25	0.53	0.52	1.01	0.83	0.22
120s horizon ($n = 20$)								
Causal-Forcing++ (2-step)	matched	9.65	8.81	0.89	0.82	0.93	0.58	0.28
Causal-Forcing++ (1-step)	matched	7.26	9.39	0.84	0.66	0.86	0.62	0.31
Causal-Forcing (frame-wise)	matched	19.9	52.7	4.82	5.06	1.08	0.67	0.24
Causal-Forcing++ (2-step, frame-wise)	native	11.4	17.3	1.89	1.54	1.01	0.75	0.09
Self-Forcing	matched	23.0	10.3	2.41	0.76	0.45	0.66	0.26
CausVid	matched	19.7	6.04	1.15	0.83	0.67	0.48	0.14
Causal-Forcing++ (1-step, frame-wise)	native	13.6	7.39	1.01	0.35	0.64	0.77	0.33
240s horizon ($n = 5$)								
Causal-Forcing++ (2-step)	matched	10.4	4.83	0.53	0.59	1.05	0.55	0.15
Causal-Forcing++ (1-step)	matched	6.42	6.66	0.56	0.51	0.83	0.67	0.30
Causal-Forcing (frame-wise)	matched	19.8	31.5	3.08	2.28	0.74	0.69	0.45
Causal-Forcing++ (2-step, frame-wise)	native	6.32	3.95	1.23	0.68	0.67	0.83	0.02
Self-Forcing	matched	22.3	5.74	1.48	1.59	0.81	0.71	0.08
CausVid	matched	18.4	2.08	0.81	0.79	0.86	0.44	0.11

Table 6. Complete I2V audit, all four horizons. The main paper reports the 60s and 240s panels; the 5s tier is an initialisation check and 120s an intermediate diagnostic. Every row is computed under the same frozen specification, so panels are comparable within a horizon; prompt sets differ across horizons, so columns are not comparable between panels. n is stated per horizon and is common to every row of that panel.

Status. Both arms are reported as preliminary and neither supports an agreement claim. What the pilot does establish is that the instrument runs end to end, that raters engage with the axis questions rather than defaulting to a side, and that the binding constraint on a powered study is pair construction rather than rater count: axis-isolated pairs, a near-tie stratum, and raters recruited outside the authors. Protocol, pair manifest and judge prompt are released verbatim so the check can be repeated and enlarged.

13. Additional Qualitative Material

References

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System	prompt mean	rank	macro mean	rank
NBF				
Infinite-Forcing	3.65	1	4.12	1
Reward-Forcing	8.51	2	10.11	4 [†]
CausVid	9.27	3	9.93	3
Rolling-Forcing	11.23	4	11.32	5 [†]
Self-Forcing	12.72	5	8.98	2 [†]
LongLive	13.86	6	14.69	6
Causal-Forcing	87.62	7	61.70	7
fBD				
Infinite-Forcing	7.75	1	6.10	1
CausVid	8.08	2	9.51	3 [†]
Self-Forcing	11.37	3	9.27	2 [†]
Reward-Forcing	14.85	4	15.61	5 [†]
LongLive	15.94	5	14.93	4 [†]
Rolling-Forcing	16.77	6	18.85	6
Causal-Forcing	22.44	7	23.49	7

Table 7. Sensitivity of the audit to category aggregation. Prompt means, as reported in the main paper, against category macro-averages, for the two static-fidelity factors at the principal horizon. [†] marks a system whose rank changes. The best- and worst-ranked systems are identical under both, so the paper’s headline contrast does not depend on the choice; intermediate positions do change, which is why no ordinal claim is made about the middle of the field.

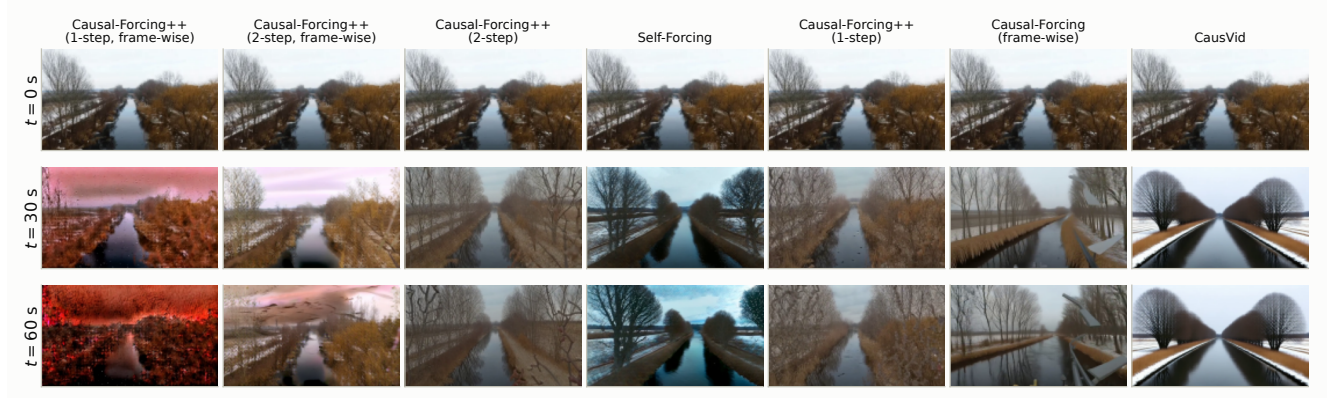


Figure 3. Image-conditioned systems on one shared source image. Every system begins from the identical frame, so divergence over the rollout is attributable to the system rather than to a differently imagined scene—which is not true of the text-conditioned track, where each system authors its own layout. Columns are ordered by increasing static-region drift.

generic metric	fBD	NBF	MCFF	DLR
VB-DD	+0.75	+0.93	+0.89	+0.36
VB-bg	-0.64	-0.71	-0.75	-0.50
VB-smooth / VB-flick	-0.96	-0.82	-0.61	-0.64

Table 8. Rank disagreement at 60s. Spearman correlation over $n = 7$ method means—descriptive, not inferential. Positive correlation between apparent motion and back-ground flow is the effect the benchmark isolates. Smoothness and flickering rank these systems identically and share a row.

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Method	Generic evidence	SNF-Bench evidence	Interpretation
CausVid	VB-DD rank 6/7	NBF rank 3/7, MCFF-L rank 4/7	stable sequence with decaying intended motion
Causal-Forcing	VB-DD rank 1/7	NBF rank 7/7, MCFF-L rank 1/7	substantial apparent motion, strongly contaminated by background displacement
Infinite-Forcing	VB-DD rank 7/7	NBF rank 1/7, MCFF-L rank 7/7	stable sequence with decaying intended motion
LongLive	VB-DD rank 3/7	NBF rank 6/7, MCFF-L rank 3/7	substantial apparent motion, strongly contaminated by background displacement
Reward-Forcing	VB-DD rank 5/7	NBF rank 2/7, MCFF-L rank 6/7	stable sequence with decaying intended motion
Self-Forcing	VB-DD rank 2/7	NBF rank 5/7, MCFF-L rank 2/7	substantial apparent motion, strongly contaminated by background displacement

Table 9. Interpretation changes between generic whole-frame metrics and SNF-Bench at 60s. Neither evaluation is labelled correct; the table identifies information hidden by whole-frame aggregation. Cases are selected by a fixed rank-gap rule applied to all public methods, so none can be selectively omitted.

Checkpoint	horizon	Δ fBD	Δ NBF	Δ MCFF-L	Δ DLR
Causal-Forcing++ (2-step)	5s	+2.47	-0.61	-0.17	-0.01
Causal-Forcing++ (2-step)	60s	-0.76	-0.95	-0.17	-0.08
Causal-Forcing++ (2-step)	120s	-1.71	-8.46	-0.73	-0.17
Causal-Forcing++ (2-step)	240s	+4.06	+0.88	-0.09	-0.28
Causal-Forcing++ (1-step)	60s	-5.05	+1.80	+0.05	-0.20
Causal-Forcing++ (1-step)	120s	-6.34	+2.00	+0.31	-0.14

Table 10. Deployment sensitivity. Change from each checkpoint’s native configuration to the common long-horizon wrapper, for the two checkpoints evaluated both ways. Absolute matched scores are never read as the published method’s performance.

- 296 prehensve benchmark for compositional text-to-video
 297 generation. In CVPR, pages 8406–8416, 2025. 5
 298 [8] Zachary Teed and Jia Deng. RAFT: Recurrent all-pairs
 299 field transforms for optical flow. In ECCV, pages 402–
 300 419, 2020. 1

Method	track	dur	n	negative	rate	most negative
CausVid	i2v	120s	20	1	5.0%	-0.055
CausVid	i2v	240s	5	0	0.0%	0.041
CausVid	i2v	5s	10	5	50.0%	-1.076
CausVid	i2v	60s	30	2	6.7%	-0.084
Causal-Forcing++ (1-step)	i2v	120s	20	1	5.0%	-0.027
Causal-Forcing++ (1-step)	i2v	240s	5	0	0.0%	0.175
Causal-Forcing++ (1-step)	i2v	5s	10	0	0.0%	0.055
Causal-Forcing++ (1-step)	i2v	60s	30	0	0.0%	0.013
Causal-Forcing (frame-wise)	i2v	120s	20	3	15.0%	-0.171
Causal-Forcing (frame-wise)	i2v	240s	5	0	0.0%	0.206
Causal-Forcing (frame-wise)	i2v	5s	10	0	0.0%	0.067
Causal-Forcing (frame-wise)	i2v	60s	30	0	0.0%	0.008
Causal-Forcing++ (2-step)	i2v	120s	20	0	0.0%	0.016
Causal-Forcing++ (2-step)	i2v	240s	5	0	0.0%	0.109
Causal-Forcing++ (2-step)	i2v	5s	10	1	10.0%	-0.164
Causal-Forcing++ (2-step)	i2v	60s	30	0	0.0%	0.008
Causal-Forcing++ (1-step, frame-wise)	i2v	120s	20	0	0.0%	0.051
Causal-Forcing++ (1-step, frame-wise)	i2v	60s	30	3	10.0%	-1.420
Causal-Forcing++ (2-step, frame-wise)	i2v	120s	20	5	25.0%	-0.241
Causal-Forcing++ (2-step, frame-wise)	i2v	240s	5	1	20.0%	-0.651
Causal-Forcing++ (2-step, frame-wise)	i2v	5s	10	1	10.0%	-0.288
Causal-Forcing++ (2-step, frame-wise)	i2v	60s	30	1	3.3%	-0.001
LTX-Video 13B-0.9.8-distilled	i2v	5s	10	2	20.0%	-0.018
Self-Forcing	i2v	120s	20	2	10.0%	-0.046
Self-Forcing	i2v	240s	5	1	20.0%	-0.136
Self-Forcing	i2v	5s	10	0	0.0%	0.064
Self-Forcing	i2v	60s	30	4	13.3%	-0.578
Wan2.1-I2V-14B-480P	i2v	5s	10	1	10.0%	-0.255
Wan2.2-I2V-A14B	i2v	5s	10	1	10.0%	-0.335
CausVid	t2v	120s	6	0	0.0%	0.047
CausVid	t2v	240s	4	1	25.0%	-0.384
CausVid	t2v	5s	6	2	33.3%	-0.093
CausVid	t2v	60s	23	0	0.0%	0.005
Causal-Forcing	t2v	120s	6	0	0.0%	0.175
Causal-Forcing	t2v	240s	4	0	0.0%	0.158
Causal-Forcing	t2v	5s	6	2	33.3%	-0.273
Causal-Forcing	t2v	60s	23	3	13.0%	-0.640
Infinite-Forcing	t2v	120s	6	0	0.0%	0.023
Infinite-Forcing	t2v	240s	4	1	25.0%	-0.158
Infinite-Forcing	t2v	5s	6	0	0.0%	0.031
Infinite-Forcing	t2v	60s	23	4	17.4%	-0.359
LongLive	t2v	120s	6	1	16.7%	-1.076
LongLive	t2v	240s	4	0	0.0%	0.022
LongLive	t2v	5s	6	1	16.7%	-0.139
LongLive	t2v	60s	23	2	8.7%	-0.051
Reward-Forcing	t2v	120s	6	1	16.7%	-0.116
Reward-Forcing	t2v	240s	4	0	0.0%	0.002
Reward-Forcing	t2v	5s	12	2	16.7%	-0.145
Reward-Forcing	t2v	60s	23	3	13.0%	-0.317
Rolling-Forcing	t2v	120s	6	3	50.0%	-0.353
Rolling-Forcing	t2v	240s	4	0	0.0%	0.043
Rolling-Forcing	t2v	5s	6	1	16.7%	-0.068
Rolling-Forcing	t2v	60s	24	2	8.3%	-0.038
Self-Forcing	t2v	120s	6	2	33.3%	-0.130
Self-Forcing	t2v	240s	4	1	25.0%	-0.153
Self-Forcing	t2v	5s	6	1	16.7%	-0.046
Self-Forcing	t2v	60s	23	4	17.4%	-1.117

Table 11. Incidence of negative DAR, i.e. clips where global-motion compensation increased measured dynamic-region flow.

category	definition
river_stream	channel water: rivers, streams, creeks, canals, waterfalls, rapids, floods
ocean_waves	open water with wave action: surf, tides, coastal swell, seascapes
precipitation	falling water or ice: rain, snowfall, and the surfaces they wet
fire_smoke	combustion plumes: wildfire, campfire, hearth, urban fire, grill smoke
lava_volcanic	volcanic flow and ejecta: lava channels, ash columns, eruption smoke
windborne	air-carried particulates and wind-driven vegetation: dust, sand, blossom, leaves, thrashing trees, drifting cloud
track/duration	n
i2v/5s	10
i2v/60s	30
i2v/120s	20
i2v/240s	5
t2v/5s	6
t2v/60s	23
t2v/120s	6
t2v/240s	4
track	source
i2v	meta
t2v	keyword
t2v	keyword-body
t2v	override
track	duration
t2v	60s

Table 12. Scene-category balance of the evaluation set, by track and horizon.

Axis	Judge	agrees with factor	called indistinguishable	pairs	inter-rater α
Static stability (fBD)	human	– (3 decided)	–	8	0.18
	vision–language	– (1 decided)	1/1	7	–
Motion persistence (MCFF-L)	human	80% [40, 100]	–	8	0.24
	vision–language	– (2 decided)	–	3	–

Table 13. Preliminary perceptual validation of the factor axes. Each judge answers one question per axis and never which output is better. agree is agreement with the factor’s ordering on pairs the factor separates clearly, over pairs the judge decided; can’t tell is the share of near-tie pairs called indistinguishable, which is the correct answer there. The unit is the pair, majority-voted across raters, and the interval is a percentile bootstrap resampling pairs rather than responses, which are clustered within rater and pair. A dash marks a cell with too few decided pairs to carry a rate; the count is given instead. This is a preliminary check on a measurement instrument, not a powered study, and no system is ranked by it.

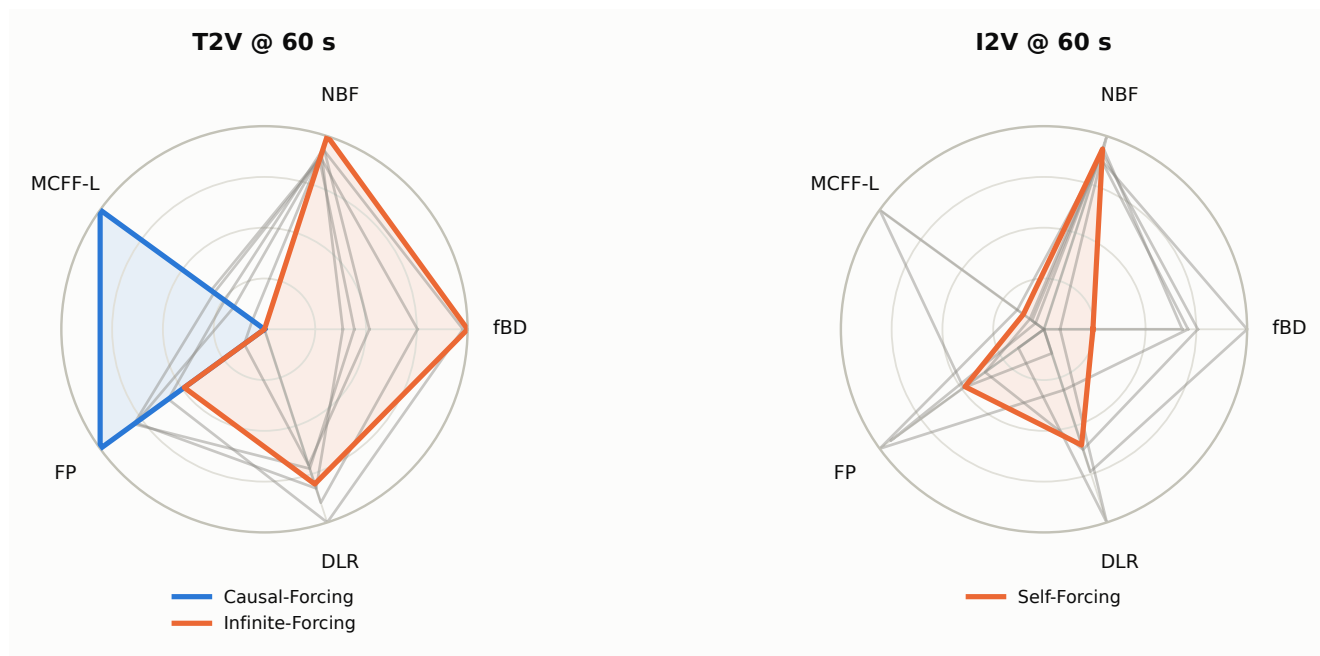


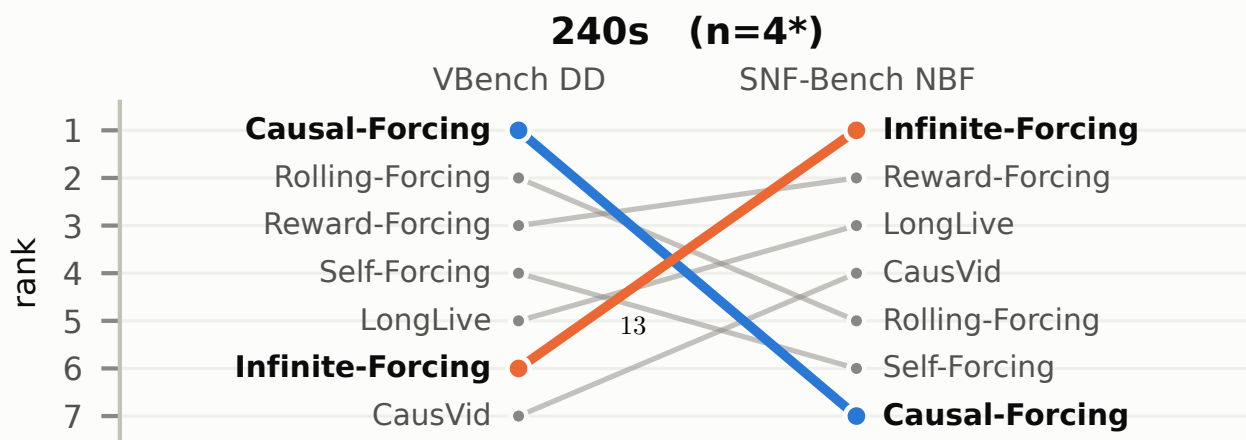
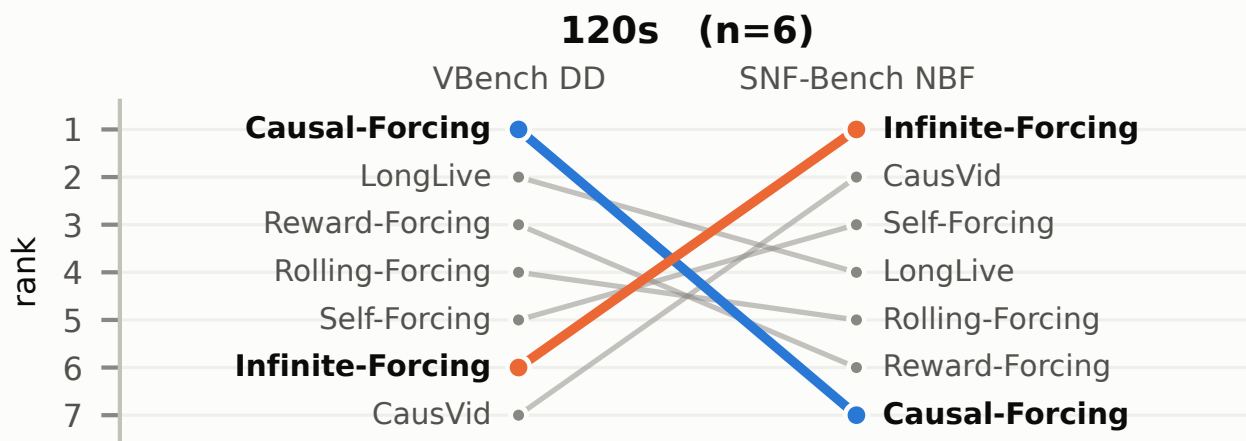
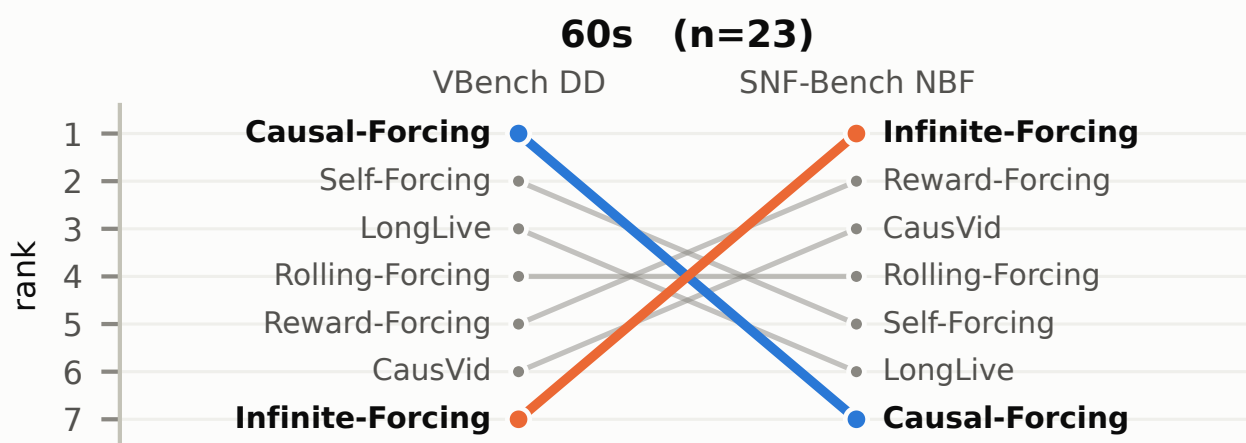
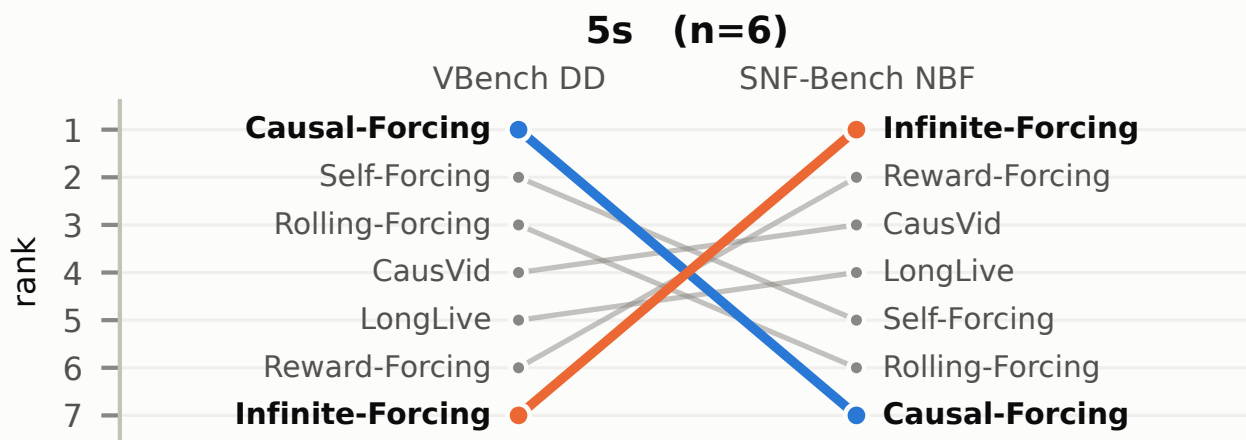
Figure 4. Factor profile per track. Each factor is rescaled across the audited systems and oriented so that outward is always better; plotting raw values would place “most drift” and “most surviving motion” both outward and the shape would carry no meaning. No system encloses the others on either track, which is the conclusion the operating-regime view reaches, shown as a profile.



Figure 5. Stated scope boundaries. (a) A two-way partition cannot separate precipitation from the support it crosses. (b) A system can attain near-best static fidelity precisely because its intended motion has also stopped. (c) High leakage with low attenuation identifies support that moves without translating, which a rigid global model does not explain.



Figure 6. Every audited text-conditioned system on one prompt. Columns are ordered by increasing static-region drift; the two highlighted systems are the pair a whole-frame motion score cannot separate.



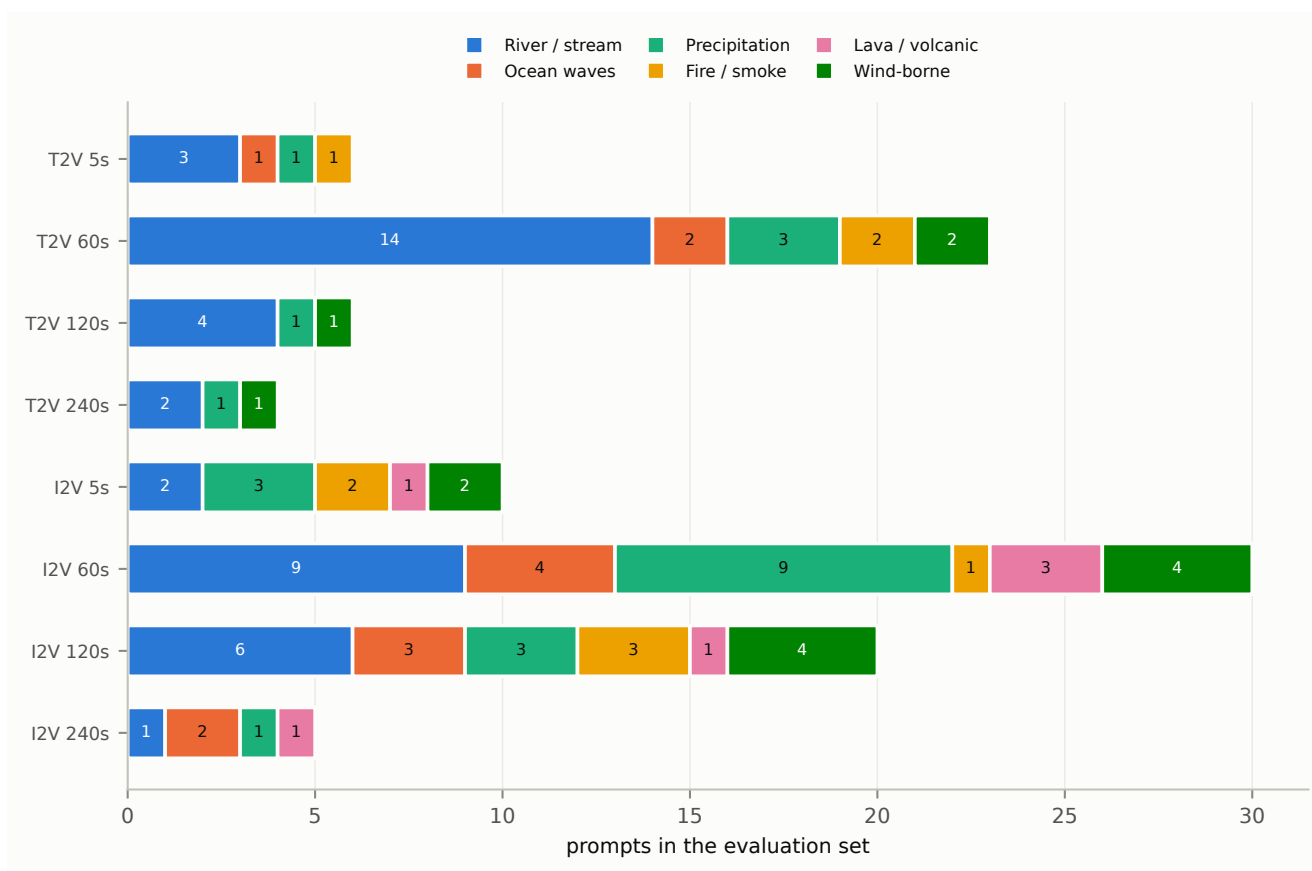


Figure 8. Composition of the evaluation set by scene category, in prompts per category at each track and horizon.